

Linear Regression

Independent variable - X	Dependent variable - Y
1	2
2	4
3	5
4	4
5	5

Mean - 3

Mean - 4

$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
-2	-2	4	4
-1	0	1	0
0	1	0	0
1	0	1	0
2	1	4	2
		10	6

Line equation = $\hat{y} = b_0 + b_1 x$
 or
 $y = mx + c$

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{6}{10} = 0.6 //$$

Using Mean

$$4 = b_0 + 0.6(3)$$

$$4 = b_0 + 1.8$$

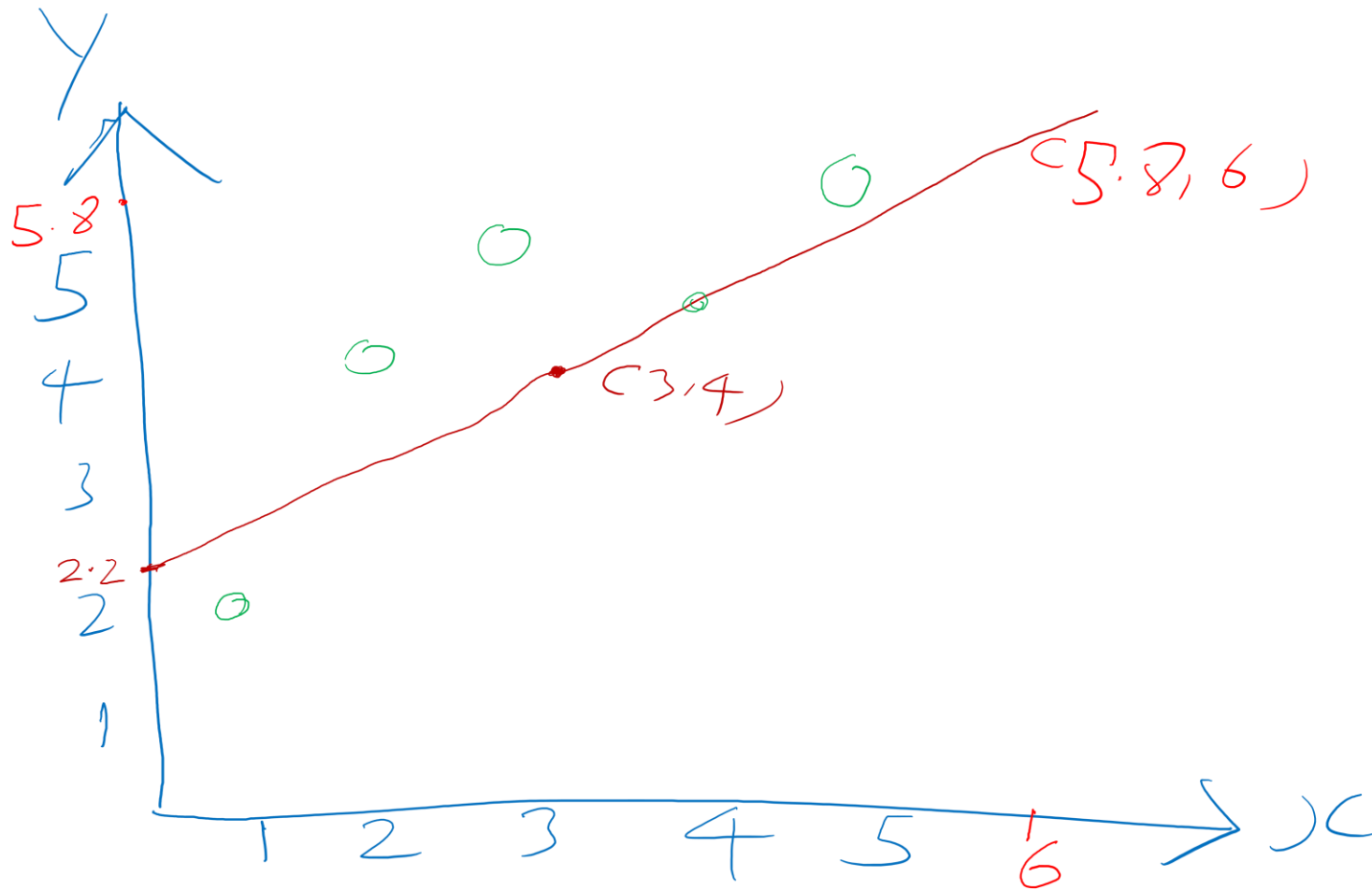
$$b_0 = 2.2 //$$

Dependent Variable

Independent Variable

$$\hat{y} = 2.2 + 0.6x$$

$y = 2.2 + 0.6x$
We want to predict
 $x = 6$
 $y = 2.2 + 0.6(6)$
 $y = 5.8$



Mean Square Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Formula

Actual value $y = [2, 4, 5, 4, 5]$

Predicted value $\hat{y} = [2.8, 3.4, 4.0, 4.6, 5.2]$

① Calculate the errors

$$\text{Errors} = [2 - 2.8, 4 - 3.4, 5 - 4.0, 4 - 4.6, 5 - 5.2]$$

$$\text{Errors} = [0.8, 0.6, -1, 0.6, 0.2]$$

(2) Square the errors

$$\begin{aligned}\text{Squared errors} &= [0.8^2 + 0.6^2, -1^2, 0.6^2, 0.2^2] \\ &= [0.64, 0.36, 1, 0.36, 0.04]\end{aligned}$$

(3) Calculate the Mean square error (MSE)

$$\begin{aligned}\text{MSE} &= \frac{1}{5} (0.64 + 0.36 + 1 + 0.36 + 0.04) \\ &= \frac{2.4}{5} = \boxed{0.48} //\end{aligned}$$

